

SUMO

From Tachiai to Triumph: A Statistical Analysis of Ring Position in Traditional Sumo

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STOR 538

07.28.2026

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Intro to Grand Sumo

- **Japan's National Sport:** Professional sumo wrestling is one of the world's oldest continuously practiced sport.
- **The ring (dohyo):** Bouts take place within a compact 4.55 m circular ring where each wrestler is assigned to East and West side of the ring to start.
- **Initial Charge (tachiai):** Matches begin with a full-contact collision where both wrestlers spring up from a low crouched position.
- **Objective:** Wrestlers aim to gain control by securing the opponents belt, applying thrusting pressure, or forcing their opponent towards the ring perimeter.
- **Win Condition:** A wrestler loses the bout immediately if any part of their body touches the ground inside or outside the ring.

Can a wrestler's position three seconds after the tachiai predict the outcome of a sumo bout?

- Early momentum can shape the rest of the bout
- Coaches emphasize gaining early ring control immediately after the tachiai
- We tested if early advantage actually predicts victory

Tachiai



3 Seconds



Early Ring
Position



Can We
Predict
the Winner?

Data Collection

- **Observational Footage:** Video footage was analyzed from official broadcasting coverage and media archives hosted on YouTube and Yahoo! Japan Sports.
- **Baseline Data:** Wrestler body weights and official divisional ranks were verified from published records on SumoStats (sumostats.com) and SumoDB (sumodb.sumogames.de) to calculate weight differentials.
- **Measurement standardization:** To ensure consistent logging across all group members, we created a standardized spreadsheet with variable definitions prior to reviewing footage.

Key Variables

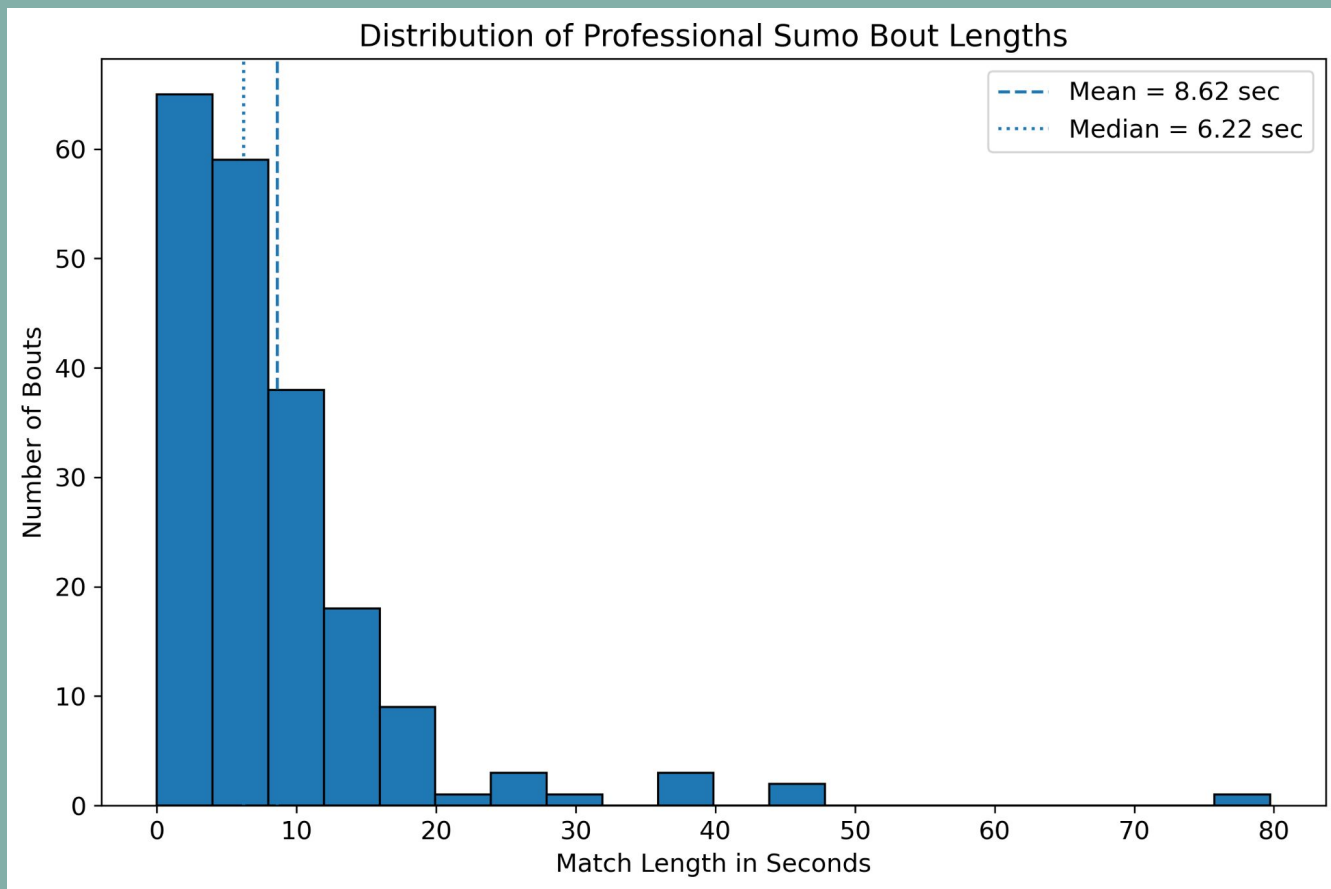
- **Primary Outcome Variable (Winner):** Categorical outcome indicating the winner. (E= East, W=West)
- **Predictor Variables:**
 - **Position at 3 Seconds:** Observed spatial location at exactly 3 seconds post-tachiai.
 - C (Center), E (Edge), O (Out)
 - **Weight Diff:** Absolute weight difference between the competing wrestlers (kg).
 - **Initial Attack:** Opening technique used at tachiai.
 - P (Push), C (Charge), B (Belt), PD (Pull down), T (Throw), S (Side Step)
 - **Match Length:** Total match duration from tachiai contact to referee ruling (sec).
 - **Winning Technique:** Official winning technique used to end the bout.
 - PO (Push out), FO (Force out), T (Throw), PD (Pull down), SO (Step out), N (No Show/ Forfeit)
 - **Rank:** Official rank titles prior to the bout.

Building the Dataset

Table 1: Preview of Sumo Observational Dataset

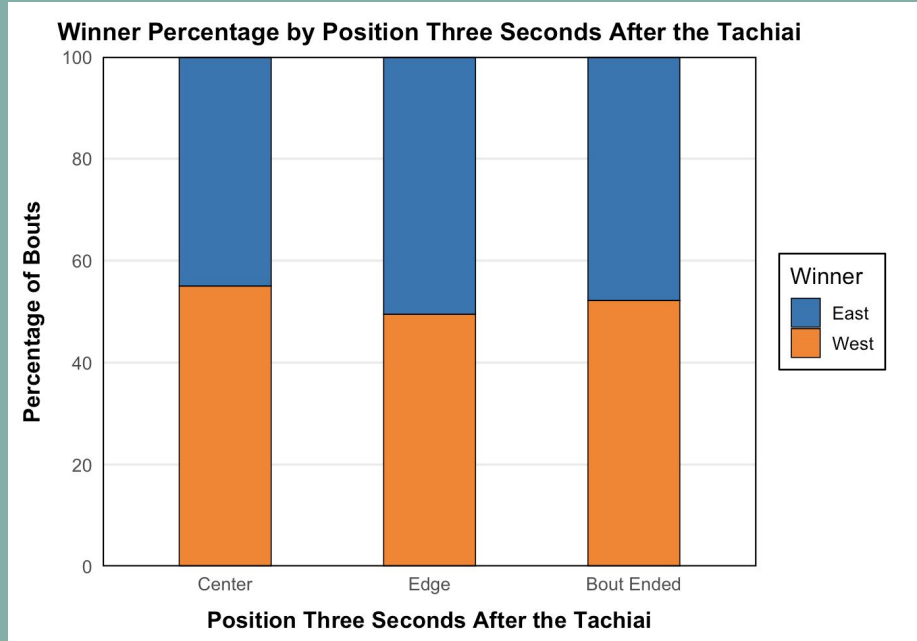
Bout #	Division	Tournament	Wrestler (E)	Rank (E)	Weight E (kg)	Wrestler W	Rank W	Weight W (kg)	Weight Diff (kg)	Initial Attack (E)	Initial Attack (W)	Position at 3 Seconds	Match Length (sec)	Winner	Winning Technique (Kimarite)
1	Jonokuchi	Hatsu Basho	Arise	Jk18e	93	Wakasasaki	Jk18w	112	19	C	C	E	3.06	E	PO
2	Jonokuchi	Hatsu Basho	Tsuyasato	Jk16w	155	Ikazuchihiyate	Jk15w	103	52	C	C	O	2.49	W	PD
3	Jonokuchi	Hatsu Basho	Goseimaru	Jk14e	118	Okada	Jk14w	164	46	C	C	O	2.78	W	PO
4	Jonokuchi	Hatsu Basho	Sawaisamu	Jk12e	106	Hara	Jk11w	110	4	C	C	C	8.78	E	PO
5	Jonokuchi	Hatsu Basho	Kyonosato	Jk9e	166	Moriurara	Jk8w	136	30	B	P	C	9.66	E	FO

- 200 traditional sumo bouts from competition footage from 2022 → 2025 across different divisions
- 1 row = 1 bout = 1 observation
- 170 unique wrestlers



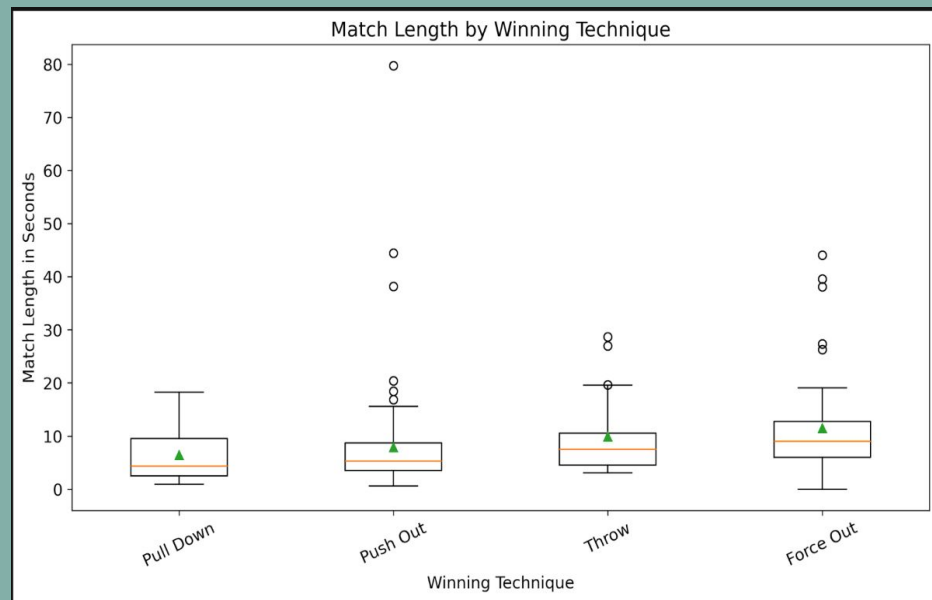
Does Position Matter?

- Center position produced nearly balanced win rates
- Outcome changed slightly being near the edge
- Position shows weak separation between winners and losers



Additional Factors

- Push outs end quickest
- Throws last longer
- Match execution matters beyond positioning



- Logistic regression
- Baseline model - signed weight diff.
- Extended model - weight difference + position
- Result: position and weight diff. + position insignificant

Model	Variable	Coefficient	Lower 95% CI	Upper 95% CI	p-value
Weight Difference Only	Intercept	-0.0735	-0.372	0.225	0.629
Weight Difference Only	Signed Weight Difference	0.0063	-0.003	0.016	0.183

Model	Variable	Coefficient	Lower 95% CI	Upper 95% CI	p-value
Weight Difference + Position	Intercept	-0.2024	-0.646	0.241	0.371
Weight Difference + Position	Position at 3 Seconds	0.2369	-0.363	0.837	0.439
Weight Difference + Position	Signed Weight Difference	0.0064	-0.003	0.016	0.176

Model Comparison

- Similar...
 - AIC
 - Log-likelihood
 - McFadden Pseudo R-squared
- Near identical model performance

Model	AIC	Log-Likelihood	McFadden Pseudo R ²
Weight Difference Only	244.509	-120.255	0.007
Weight Difference + Position	245.91	-119.955	0.01

Coaching Takeaway

- Use early position as context, rather than prediction
- Technique & execution matter more than location alone
- Combine video review w/ multiple performance indicators



Critiques and Limitations

- **Omission of Belt Control:**
 - Recorded spatial position (Position at 3 Seconds) and Initial Attack but omitted whether a wrestler secured a belt grip.
 - Belt control provides offensive leverage even when forced back to the edge which could have masked the true predictive effect of ring position.
- **Role-Based Tracking:**
 - Spatial location was tracked as a general ring status (Center vs Edge) rather than directional momentum; future studies should categorize position by role (Attacker at Edge vs Defender at Edge) to measure true positional leverage.
- **Competition Rank Disparity:**
 - While the mixed model controlled for division-level baseline differences, it did not account for individual rank gaps.

Citations

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